

Introduction

A computer is an electronic device that processes data to produce meaningful information.

It is made up of two main parts:

- **Hardware:** The physical parts of the computer such as CPU, RAM, motherboard, etc.
- **Software:** The set of instructions that tells the hardware what to do.

Understanding how to assemble, connect, and troubleshoot computer hardware is a basic but vital skill for every engineering student.

Major Components and Their Purpose

Component	Purpose	Working Principle
Motherboard	Main printed circuit board connecting all components.	Acts as the communication backbone allowing CPU, memory, storage, and peripherals to exchange data.
Processor (CPU)	Executes all instructions and performs calculations.	Fetches, decodes, and executes instructions from memory and controls all computer operations.
RAM (Memory)	Temporarily stores active data and programs for quick access.	Volatile memory that provides extremely fast read/write access for the CPU.
HDD (Hard Disk Drive)	Permanent magnetic storage for OS and user data.	Uses spinning platters and magnetic heads to read and write data.
SSD (Solid State Drive)	Permanent storage using flash memory chips.	Stores data electronically with no moving parts, providing faster access than HDD.
CMOS Battery	Maintains BIOS settings and real-time clock when system is off.	Powers the BIOS memory chip to retain date, time, and hardware configuration.

Component	Purpose	Working Principle
SMPS (Power Supply Unit)	Converts AC power to DC power and distributes it to components.	Switches and regulates power to provide stable DC voltages to motherboard and peripherals.

Power Supply Unit (PSU)

The **Power Supply Unit (PSU)** is a hardware component that **converts electrical energy from the wall outlet (AC – alternating current) into usable DC (direct current) power** for the computer's internal components.

Without the PSU, the computer cannot operate, because components like the CPU, motherboard, RAM, and drives require stable DC voltage to function.

Purpose / Importance

- Provides **stable and regulated DC power** to all components.
- Protects hardware from **voltage fluctuations, surges, or power spikes**.
- Supplies the **correct voltage and current** required by each component.

Why PSU is needed:

- **Motherboard & CPU:** Need 12V, 5V, and 3.3V DC for operation.
- **HDD / SSD / Optical Drives:** Usually need 5V or 12V DC.
- **Fans and peripherals:** Need low-voltage DC for proper operation.

Without a PSU, no component will work. A faulty PSU can also **damage sensitive components**.

How It Works

The PSU converts AC from the wall outlet (usually 230V in India, 110V in the USA) into several DC voltages required by the computer. It does this in a few stages:

1. **AC Input:** The PSU receives AC voltage from the mains.

2. **Rectification:** Converts AC to DC using diodes.
3. **Filtering:** Smoothens the DC to remove ripples.
4. **Regulation:** Ensures stable voltage output to prevent damage to components.
5. **Distribution:** Sends appropriate voltages (12V, 5V, 3.3V) to the motherboard, drives, and peripherals.

Modern PSUs also include:

- **Over-voltage protection**
- **Over-current protection**
- **Short-circuit protection**

Types of PSU

1. **ATX PSU** – Standard desktop PSU for most computers.
2. **SFX PSU** – Small form factor PSU for compact PCs.
3. **Modular PSU** – Allows users to connect only the cables they need.

Common Specifications

- **Wattage:** Total power the PSU can supply, e.g., 450W, 650W, 850W.
- **Efficiency:** How effectively it converts AC to DC (80%, 80 Plus Bronze, Silver, Gold).
- **Rails:** 12V rail(s) are most important for CPU and GPU power.

Signs of PSU Failure

- Computer doesn't turn on.
- Random shutdowns or restarts.
- Burning smell or unusual noise.
- Voltage fluctuations affecting performance.

Motherboard

The **Motherboard** (also called the mainboard or system board) is the **primary printed circuit board (PCB)** in a computer that connects and allows communication between all hardware components, including the CPU, RAM, storage devices, power supply, and peripheral devices.

It acts as the **backbone of the computer**, providing pathways (buses) for data, power, and signals to flow between components.

Purpose / Importance

- **Central Hub:** Connects all major components of a computer.
- **Communication:** Enables CPU, memory, storage, and peripherals to communicate.
- **Power Distribution:** Routes DC power from the PSU to components.
- **Expansion:** Provides slots and ports for additional devices (graphics cards, sound cards, network cards).
- **BIOS/UEFI Hosting:** Houses firmware that controls boot-up and hardware initialization.

Without a motherboard, **individual components cannot work together** as a functional system.

Major Components of a Motherboard

Component	Function
CPU Socket	Slot where the processor is installed. Supports specific CPU types.
RAM Slots (DIMM slots)	Slots for RAM modules to provide temporary memory for the CPU.
Chipset	Controls communication between CPU, memory, and peripherals. Determines features and performance.
Power Connectors	Connects PSU to provide power to the motherboard and components.
Expansion Slots (PCIe, PCI)	For installing additional cards (GPU, sound card, network card).
Storage Connectors (SATA / M.2)	Connects HDDs, SSDs, and optical drives.
USB / Front Panel Headers	Connects USB ports, power/reset buttons, LEDs, and audio jacks.
CMOS Battery	Powers BIOS settings when system is off.
VRM (Voltage Regulator Module)	Provides stable power to CPU and other components.

How the Motherboard Works

1. **Power Distribution:** Receives DC power from PSU and distributes it to CPU, RAM, drives, and other components.
2. **Data Communication:**

- Uses **buses** and **chipsets** to transfer data between CPU, RAM, storage, and peripherals.
- PCIe slots provide high-speed communication with graphics cards and NVMe drives.

3. **Boot Process:**

- BIOS/UEFI firmware initializes components, performs POST (Power-On Self-Test), and loads the OS from storage into RAM.

4. **Peripheral Support:**

- USB, audio, network, and other ports allow external devices to interact with the system.

Types of Motherboards

- **ATX:** Standard full-size board for desktops.
- **Micro-ATX:** Smaller form factor, fewer slots but compatible with ATX cases.
- **Mini-ITX:** Compact boards for small form factor PCs.

Each type has different **number of slots, ports, and expansion options**.

Signs of Motherboard Problems

- Computer fails to boot or POST.
- No display output even though CPU and RAM are fine.
- USB or peripheral devices not detected.
- Random shutdowns or instability.

CPU (Central Processing Unit)

The **CPU**, often called the **brain of the computer**, is the component that **executes instructions** from programs and performs **calculations, logic operations, and data processing**.

Without the CPU, a computer cannot perform any computation or run software.

Purpose / Importance

- **Processing Data:** Executes program instructions and handles computations.
- **Controlling Operations:** Directs the flow of data between memory, storage, and input/output devices.

- **Task Execution:** Handles multitasking by rapidly switching between processes.

Why CPU is needed:

Every program you run—whether it's a browser, game, or office application—relies on the CPU to process instructions and produce results.

Major Components of a CPU

Component	Function
ALU (Arithmetic Logic Unit)	Performs all arithmetic (addition, subtraction) and logical (AND, OR, comparisons) operations.
Control Unit (CU)	Directs the flow of data between CPU, RAM, and peripherals. Interprets instructions from programs.
Registers	Small, ultra-fast memory locations inside CPU for temporary storage of data and instructions.
Cache Memory (L1, L2, L3)	Extremely fast memory built into or near the CPU to reduce access time to frequently used data.

How the CPU Works

The CPU works in a **cycle of fetching, decoding, and executing instructions**, known as the **Instruction Cycle**:

1. **Fetch:** CPU fetches the instruction from RAM.
2. **Decode:** Control Unit interprets the instruction to understand what operation is required.
3. **Execute:** ALU performs the operation (calculation or logical decision).
4. **Store:** Results are written back to RAM or registers for further use.

Modern CPUs can perform **billions of instructions per second (measured in GHz)** and often have **multiple cores** to handle parallel processing.

CPU Types

- **Desktop CPUs:** Intel Core (i3, i5, i7, i9), AMD Ryzen (3, 5, 7, 9)
- **Server CPUs:** Intel Xeon, AMD EPYC
- **Mobile CPUs:** Intel Core Mobile, ARM processors for laptops and smartphones

Signs of CPU Problems

- Computer does not boot.
- System freezes or crashes during intensive tasks.
- Overheating or fan running constantly.
- Beep codes during POST indicating CPU error.

RAM (Random Access Memory)

RAM is a type of **volatile memory** that temporarily stores data and instructions that the CPU needs **while executing programs**. Volatile means that all stored data is **lost when the computer is powered off**.

RAM allows the CPU to access data **much faster than from permanent storage (HDD or SSD)**.

Purpose / Importance

- Provides **fast temporary storage** for running programs and active processes.
- Improves system **multitasking** and responsiveness.
- Stores **data that CPU needs immediately**, such as program instructions, variables, and files currently in use.

Without RAM:

- Programs cannot run efficiently.
- CPU would have to access data directly from HDD/SSD, which is **much slower**.

Types of RAM

Type	Key Features	Usage
DRAM (Dynamic RAM)	Needs constant refreshing; slower than SRAM	Main system memory
SRAM (Static RAM)	Faster and more expensive; doesn't need refreshing	CPU cache memory
SDRAM (Synchronous DRAM)	Synchronized with CPU clock for faster access	Modern desktops

Type	Key Features	Usage
DDR1 / DDR2 / DDR3 / DDR4 / DDR5	Successive generations of SDRAM with higher speed and efficiency	Desktop/laptop memory

DDR (Double Data Rate) indicates **data is transferred twice per clock cycle**, increasing speed with each generation.

How RAM Works

1. When a program is executed, it is loaded from **HDD/SSD into RAM**.
2. CPU fetches instructions and data **directly from RAM**, because it is much faster than storage.
3. CPU processes the data and temporarily stores results in RAM.
4. Once the computer is shut down, all RAM contents are **erased**.

RAM vs HDD/SSD

Feature	RAM	HDD	SSD
Speed	Fastest (~10-100 ns)	Slow (~5-15 ms)	Fast (~50-150 μ s)
Volatility	Volatile	Non-volatile	Non-volatile
Purpose	Temporary storage for active programs	Long-term storage	Long-term storage, faster than HDD

Signs of RAM Problems

- System fails to boot or beeps during POST.
- Random crashes or application freezes.
- Blue Screen of Death (BSOD) in Windows.
- Data corruption or programs not opening.

HDD (Hard Disk Drive)

An **HDD** is a type of **permanent storage device** that uses **magnetic storage** to store digital data. Unlike RAM, it **retains data even when the computer is powered off** (non-volatile memory).

HDDs are commonly used for storing the **operating system, applications, and user files**.

Purpose / Importance

- Provides **long-term storage** for files, software, and the operating system.
- Allows access to large amounts of data at a **lower cost per GB** compared to SSDs.
- Serves as **secondary storage**, working alongside RAM (temporary memory) for system operations.

Without storage drives like HDD or SSD, a computer **cannot save data permanently**.

How HDD Works

1. **Platters:** Circular disks coated with magnetic material store data in sectors and tracks.
2. **Read/Write Head:** Positioned on an actuator arm, it **magnetizes spots on the platter** to write data and **reads magnetized spots** to retrieve data.
3. **Spindle Motor:** Rotates the platters at high speed (e.g., 5400–7200 RPM for desktops).
4. **Controller Board:** Manages data transfer between the computer and the HDD.

HDD vs SSD vs RAM

Feature	HDD	SSD	RAM
Speed	Slow (ms)	Fast (μ s)	Very fast (ns)
Volatility	Non-volatile	Non-volatile	Volatile
Storage	Large capacity (TB)	Moderate capacity (GB–TB)	Small capacity (GB)
Mechanical parts	Yes (platters & heads)	No	No
Cost per GB	Low	Higher	Very high

Types of HDD

1. **Desktop HDD:** 3.5-inch platters, used in desktop PCs.
2. **Laptop HDD:** 2.5-inch platters, used in laptops.
3. **Enterprise/Server HDD:** High-speed, high-capacity drives with better durability.

Signs of HDD Problems

- System fails to boot or OS not detected.
- Clicking or grinding noises (head/platter issues).
- Files become corrupted or inaccessible.
- Slow file access or frequent freezing.

CMOS

(Complementary Metal-Oxide-Semiconductor)

CMOS refers to a small amount of **volatile memory on the motherboard** that stores **BIOS/UEFI settings** such as:

- System date and time
- Boot sequence
- Hardware configuration (CPU, RAM, drives)
- Power management settings

The memory is powered by a **small battery (CMOS battery)** so that settings are retained even when the computer is turned off.

Purpose / Importance

- **Stores BIOS/UEFI Settings:** Ensures your custom hardware configurations are retained.
- **Maintains System Clock:** Keeps track of real time even when the system is off.
- **Enables Boot Process:** BIOS reads CMOS settings to initialize hardware and load the operating system.

Without CMOS (or a functional CMOS battery):

- BIOS settings reset to defaults on every startup.
- Date and time may be incorrect.
- System may fail to boot if custom configurations are required.

How CMOS Works

1. The **BIOS firmware** reads CMOS memory during system startup.

2. CMOS memory provides the **necessary hardware configuration** to the CPU and motherboard.
3. CMOS is powered by a **small coin-cell battery**, usually **3V (CR2032)**.
4. When settings are changed (e.g., boot order), the BIOS writes them to CMOS memory.

CMOS vs BIOS

Feature	CMOS	BIOS/UEFI
Function	Stores configuration settings	Firmware that initializes hardware and boots OS
Memory Type	Volatile (battery-backed)	Non-volatile
Dependency	Requires CMOS battery to retain data	Independent of battery

Signs of CMOS/CMOS Battery Problems

- Incorrect date and time after reboot.
- BIOS settings reset automatically.
- Computer may show a **"CMOS Checksum Error"** on startup.
- System fails to retain custom boot sequence.

Why RAM is Faster than HDD and SSD

RAM is built using semiconductor circuits (DRAM or SRAM) that allow data to be accessed **directly and instantly**, without any moving parts or electronic delays caused by file structures.

Component	Typical Access Time	Type of Access	Nature
RAM	~10–100 nanoseconds	Direct (semiconductor)	Volatile
SSD	~50–150 microseconds	Flash memory (electronic)	Non-volatile
HDD	~5–15 milliseconds	Mechanical (rotating disk)	Non-volatile

Thus, RAM is around **1000× faster than SSD** and **10,000× faster than HDD**, which is why the operating system loads programs into RAM before execution.

How Components Work Individually and Together

1. **Power On:** SMPS converts AC to DC and supplies power to the motherboard.
2. **BIOS Initialization:** CMOS battery powers BIOS, which checks all components (POST – Power-On Self-Test).
3. **Boot Process:** BIOS loads the OS from HDD/SSD into RAM.
4. **Execution:** CPU fetches and executes instructions from RAM.
5. **Coordination:** The motherboard allows communication between CPU, memory, and storage.
6. **User Interaction:** Once initialized, input/output devices allow the user to operate the system.

Step-by-Step: Assembling a Computer

1. Prepare your workspace with an anti-static mat and tools.
2. Mount the **CPU** on the motherboard and apply thermal paste.
3. Install the **RAM** in its slot (ensure it clicks in properly).
4. Mount **HDD/SSD** and connect SATA data and power cables.
5. Fit the **SMPS** and connect motherboard and CPU power cables.
6. Attach front panel connectors, keyboard, mouse, and monitor.
7. Power on the system, enter BIOS, and install the operating system.

Data Flow Example – Opening a Browser

1. You double-click the browser icon.
2. The **CPU** sends a request to load program data from storage.
3. The program (browser) is fetched from **SSD/HDD** into **RAM**.
4. The **CPU** executes instructions directly from RAM.
5. When you visit a website, data passes through the **Network Interface Card (NIC)** to the internet.
6. Output is displayed on the **monitor**.

Common Hardware Problems and Identification

Symptom	Possible Cause	Solution
No display	Loose RAM or cable	Reseat RAM, check connections
Random restart	Overheating or PSU issue	Clean fan, replace SMPS
No power	Faulty SMPS or cable	Replace power cable or PSU
Continuous beeping	Hardware error	Check RAM or GPU
BIOS not saving time	Dead CMOS battery	Replace CMOS battery

Safety Precautions

- Always disconnect power before opening the case.
- Use an **anti-static wrist strap**.
- Avoid touching circuit pins or chips directly.
- Never force components into slots.
- Ensure proper cooling and ventilation.